Editors' Choice

From the Society for Vascular Surgery

The impact of revascularization strategy on clinical failure, hemodynamic failure, and chronic limb-threatening ischemia symptoms in the BEST-CLI Trial

Matthew T. Menard, MD,[a] Alik Farber, MD, MBA,[b] Gheorghe Doros, PhD, MBA,[c] Katherine L. McGinigle, MD, MPH,[d] Emiliano Chisci, MD, PhD,[e] Leonardo C. Clavijo, MD, PhD,[f] Ahmed Kayssi, MD,[g] Peter A. Schneider, MD,[h] Beau M. Hawkins, MD,[i] Michael D. Dake, MD,[j] Taye Hamza, PhD,[k] Michael B. Strong, MA,[a] Kenneth Rosenfield, MD,[l] *and* Michael S. Conte, MD,[h]

Boston and Watertown, MA; Chapel Hill, NC; San Luis Obispo and San Francisco, CA; Oklahoma City, Ok; Tucson, AZ; Florence, Italy; and Toronto, ON, Canada

ABSTRACT

Objective: Sustained clinical and hemodynamic benefit after revascularization for chronic limb-threatening ischemia (*CLTI*) is needed to resolve symptoms and prevent limb loss. We sought to compare rates of clinical and hemodynamic failure as well as resolution of initial and prevention of recurrent CLTI after endovascular (*ENDO*) vs bypass (*OPEN*) revascularization in the Best-Endovascular-versus-best-Surgical-Therapy-in-patients-with-CLTI (*BEST-CLI*) trial.

Methods: As planned secondary analyses of the BEST-CLI trial, we examined the rates of (1) clinical failure (a composite of all-cause death, above-ankle amputation, major reintervention, and degradation of WIfI stage); (2) hemodynamic failure (a composite of above-ankle amputation, major and minor reintervention to maintain index limb patency, failure to an initial increase or a subsequent decrease in ankle brachial index of 0.15 or toe brachial index of 0.10, and radiographic evidence of treatment stenosis or occlusion); (3) time to resolution of presenting CLTI symptoms; and (4) incidence of recurrent CLTI. Time-to-event analyses were performed by intention-to-treat assignment in both trial cohorts (cohort 1: suitable single segment great saphenous vein [SSGSV], N = 1434; cohort 2: lacking suitable SSGSV, N = 396), and multivariate stratified Cox regression models were created.

Results: In cohort 1, there was a significant difference in time to clinical failure (log-rank $P < .001$), hemodynamic failure (log-rank $P < .001$), and resolution of presenting symptoms (log-rank $P = .009$) in favor of OPEN. In cohort 2, there was a significantly lower rate of hemodynamic failure (log-rank $P = .006$) favoring OPEN, and no significant difference in time to clinical failure or resolution of presenting symptoms. Multivariate analysis revealed that assignment to OPEN was associated with a significantly lower risk of clinical and hemodynamic failure in both cohorts and a significantly higher likelihood of resolving initial and preventing recurrent CLTI symptoms in cohort 1, including after adjustment for key baseline patient covariates (end-stage renal disease [ESRD], prior revascularization, smoking, diabetes, age >80 years, WIfI stage, tissue loss, and infrapopliteal disease). Factors independently associated with clinical failure included age >80 years in cohort 1 and ESRD across both cohorts. ESRD was associated with hemodynamic failure in cohort 1. Factors associated with slower resolution of presenting symptoms included diabetes in cohort 1 and WIfI stage in cohort 2.

From the Division of Vascular and Endovascular Surgery, Brigham and Women's Hospital, Harvard Medical School, Boston[a]; the Division of Vascular and Endovascular Surgery, Boston Medical Center, Boston University Chobanian & Avedisian School of Medicine, Boston[b]; the School of Public Health, Boston University, Boston[c]; the Division of Vascular Surgery, School of Medicine, University of North Carolina at Chapel Hill, Chapel Hill[d]; the Department of Surgery, Vascular Surgery Division, San Giovanni di Dio Hospital, Florence[e]; the Coastal Cardiology, French Hospital Medical Center, San Luis Obispo[f]; the Division of Vascular Surgery, University of Toronto, Toronto[g]; the Division of Vascular and Endovascular Surgery, University of California, San Francisco, San Francisco[h]; the Cardiovascular Section, University of Oklahoma Health Sciences Center, Oklahoma City[i]; the Department of Medical Imaging, University of Arizona Health Sciences, Tucson[j]; the HealthCore, Inc, Watertown[k]; and the Section of Vascular Medicine and Intervention Massachusetts General Hospital, Harvard Medical School, Boston.[l]

Presented at the 2023 Society for Vascular Surgery Vascular Annual Meeting, National Harbor, MD, June 14-17, 2023.

Additional material for this article may be found online at www.jvascsurg.org.

Correspondence: Matthew T. Menard, MD, Division of Vascular and Endovascular Surgery, Brigham and Women's Hospital, 75 Francis Street, Boston, MA 02115 (e-mail: mmenard@bwh.harvard.edu).

The editors and reviewers of this article have no relevant financial relationships to disclose per the JVS policy that requires reviewers to decline review of any manuscript for which they may have a conflict of interest.

0741-5214

https://doi.org/10.1016/j.jvs.2024.07.085

Conclusions: Durable clinical and hemodynamic benefit after revascularization for CLTI is important to avoid persistent and recurrent CLTI, reinterventions, and limb loss. When compared with ENDO, initial treatment with OPEN surgical bypass, particularly with available saphenous vein, is associated with improved clinical and hemodynamic outcomes and enhanced resolution of CLTI symptoms. (J Vasc Surg 2024;80:1755-65.)

Keywords: Critical limb-threatening ischemia; Hemodynamic; Revascularization; Endovascular; Surgical bypass; Wound healing; Peripheral artery disease; WIfI

Sustained clinical and hemodynamic benefit after revascularization for chronic limb-threatening ischemia (CLTI) is necessary to achieve durable resolution of presenting symptoms and prevent limb loss. Since its initial introduction as a component of the Society for Vascular Surgery's (SVS) proposed objective performance goals for CLTI[1] in 2009, the use of major adverse limb events (MALE), comprising major amputation and major reinterventions, has gained increasing traction as an important efficacy outcome measure.[2,3]

Few if any investigators, however, have attempted to capture the impact of revascularization strategies for CLTI on either short- or long-term clinical or hemodynamic failure. We sought to compare rates of clinical and hemodynamic failure as well as resolution of initial and prevention of recurrent CLTI symptoms after endovascular or surgical revascularization in the Best-Endovascular-versus-best-Surgical-Therapy-in-patients-with-CLTI (BEST-CLI) trial. To our knowledge, this is the first report describing the direct effect of the initial treatment strategy on preventing recurrent ischemic symptoms. The BEST-CLI trial was a multicenter, open-label, randomized controlled trial comparing outcomes in patients who underwent either open surgical bypass (OPEN) or endovascular intervention (ENDO) for CLTI due to infrainguinal peripheral arterial disease.[2] The primary results of the trial have been reported elsewhere.[4] Herein we present an analysis of several prespecified novel secondary end points of the trial, including hemodynamic failure, clinical failure, resolution of initial CLTI symptoms, and the development of new or recurrent symptoms on an intention-to-treat (ITT) basis.

METHODS

The full design and protocol details of the BEST-CLI trial have been previously reported.[2] Participants provided written informed consent, and the protocol was approved by the Human Research Subjects Committee at each participating institution (ClinicalTrials.gov, NCT02060630). Bilateral assessment of the great saphenous vein allowed randomization into one of two parallel cohorts. Cohort 1 included 1434 patients with presumed adequate great saphenous vein for a single segment (SSGSV) bypass, whereas 396 patients lacking adequate SSGSV on preoperative assessment were randomized into cohort 2. The primary trial outcome was the occurrence of MALE or death from any cause. MALE included ipsilateral above-ankle amputation or major reintervention. Major reinterventions were defined in accordance with the SVS objective performance goals[1] and included thrombectomy or thrombolysis, a new open bypass procedure or a major open revision (jump or interposition graft) of an existing bypass.

As planned secondary analyses of the BEST-CLI trial, we examined the occurrence of the following end points.

- Clinical failure—defined as a composite of all-cause death, above-ankle amputation, major reintervention, and degradation of WIfI stage.
- Hemodynamic failure—defined as a composite of above-ankle amputation, any reintervention to maintain index limb patency, failure to increase the ankle brachial index (ABI) by 0.15 or toe brachial index (TBI) by 0.10 postprocedure, a subsequent decrease in the ABI of 0.15 or TBI of 0.10 compared with the 30-day postprocedure value, and/or radiographic evidence of >50% arterial stenosis or occlusion in the treated index limb.
- Time to resolution of presenting CLTI symptoms—defined as the time to resolution of presenting CLTI symptoms in the index limb. This represents the first time point at which the patient has resolved initial rest pain, healed their presenting tissue loss, or both.
- Incidence of new or recurrent CLTI—defined as new or recurrent CLTI events (ischemic rest pain or tissue loss) from the date of randomization through the end of study. Patients who failed to resolve their initial symptoms were censored with regard to this end point.

Minor reinterventions included percutaneous procedures without thrombus management (angioplasty, stenting, atherectomy) or surgical patch angioplasty. Major amputation was defined as any above-ankle amputation. Clinical status, including the presence of persistent, new, or recurrent rest pain or ulcerations at specific locations (calf, hindfoot, forefoot, or individual toes of the index limb), and hemodynamic status, including ABI and TBI, were assessed at in-person study visits at 1, 3, 6, 12, 18, 24 months and annually thereafter by a physician or advanced practice practitioner. Time to resolution of presenting symptoms was analyzed assuming different censoring scenarios, including from the date of randomization through the end of study; through the end of study or above-ankle amputation, whichever comes first; and through the end of study with all-cause death and above-ankle amputation as competing risks.

Time-to-event analyses are presented as Kaplan-Meier estimates, with log-rank comparisons by ITT assignment. Adjustments were made for predefined baseline covariates (end-stage renal disease, prior index limb revascularization, smoking, diabetes, age >80 years, WIfI stage, tissue loss, presence of significant infrapopliteal [IP] disease) known to influence treatment outcomes in CLTI. Both tissue loss (vs rest pain) and significant IP disease were predefined as stratification variables for randomization in the BEST-CLI trial. Missing baseline covariates were imputed using multiple imputation. Sensitivity analyses were performed adjusting for imputed complete baseline characteristics. Hazard ratios were generated using stratified Cox regression models. Time to resolution of presenting CLTI symptoms was analyzed both with and without all-cause death and above-ankle amputation considered as competing risks.

The incidence of new or recurrent CLTI compared by the ITT group was analyzed as person-years of follow-up incidence rates and reported as incident rate ratios. Negative binomial regression was used to compare the number of events between treatment arms. Patients failing to resolve presenting CLTI symptoms were excluded from this end point.

We examined univariate associations of defined covariates with selected outcomes using log-rank and then created Cox proportional hazards models to identify independent predictors for the entire study population and separately within each assigned treatment arm. Baseline variables examined included age, sex, diabetes, smoking, prior index limb revascularization, chronic kidney disease, WIfI (wound grade, ischemia grade, infection grade, and overall stage), tissue loss vs rest pain, IP disease, hypertension, hyperlipidemia, medications (antiplatelet, anticoagulant, and statin use), levels of disease treated (femoropopliteal [FP] only, IP only, or both FP+IP), and type of intervention used for ENDO and type of conduit used for OPEN. Factors associated with end points on univariate screen ($P \leq 2$) were selected for the multivariable models. Poisson regression was used to identify predictors of total number of events. Statistical significance was considered at a level of $P < .05$, without corrections made for multiple testing.

RESULTS

Baseline characteristics of the analyzed patient cohorts are found in Supplementary Table I (online only).

Number and type of secondary end point events

Clinical failure. In cohort 1, a total of 355 of the 718 (49%) patients assigned to OPEN experienced clinical failure during the entire follow-up period (median 2.7 years). All-cause death, above-ankle amputation, major reintervention, and degradation of WIfI stage occurred in 237, 74, 66, and 124 patients, respectively (Table I). Major reinterventions included a new open bypass graft (33), thrombectomy or thrombolysis (66), and major open revisions (21) (Supplementary Table II, online only). In comparison, 477 of the 716 (67%) patients assigned to ENDO experienced clinical failure. All-cause death, above-ankle amputation, major reintervention, and degradation of WIfI stage occurred in 269, 106, 167, and 180 patients, respectively (Table I). Major reinterventions in the ENDO group included an open bypass procedure (186), major open graft revision (7), and thrombectomy or thrombolysis (65) (Supplementary Table II, online only).

> **ARTICLE HIGHLIGHTS**
> - **Type of Research:** Prospective, multicenter randomized controlled study
> - **Key Findings:** A randomized controlled study of 1830 patients with critical limb-threatening ischemia demonstrated improved hemodynamic and clinical outcomes when treated with surgical bypass compared with endovascular therapy.
> - **Take Home Message:** Open surgical bypass, particularly when performed with available saphenous vein, was more effective at healing wounds and preventing recurrent ischemia than endovascular therapy.

In cohort 2, a total of 96 of the 197 (49%) patients assigned to OPEN experienced clinical failure during the entire follow-up period (median 1.6 years). All-cause death, above-ankle amputation, major reintervention, and degradation of WIfI stage occurred in 51, 30, 28, and 35 patients, respectively (Table I). Major reinterventions included a new open bypass graft (14), thrombectomy or thrombolysis (38), and major open revisions (5) (Supplementary Table II, online only). In comparison, 116 of the 199 (58%) patients assigned to ENDO experienced clinical failure. All-cause death, above-ankle amputation, major reintervention, and degradation of WIfI stage occurred in 48, 28, 51, and 42 patients, respectively (Table I). Major reinterventions in the ENDO group included an open bypass procedure (54), major open graft revision (3), and thrombectomy or thrombolysis (28) (Supplementary Table II, online only).

Hemodynamic failure. In cohort 1, a total of 61% and 73% patients assigned to OPEN and ENDO, respectively, experienced hemodynamic failure. Among those assigned to OPEN, 74 (10%) had an above-ankle amputation, 223 (31%) had a major and/or minor reintervention to maintain index limb patency, 179 (25%) did not initially increase the ABI by 0.15 or TBI by 0.10, 165 (23%) had a subsequent decrease in the ABI of 0.15 or TBI of 0.10, and 76 (11%) had radiographic evidence of arterial stenosis or occlusion in the treated index limb (Table II). Among those assigned to ENDO, 106 (15%) had an above-ankle

Table I. Clinical failure and component events for cohorts 1 and 2

	Cohort 1		Cohort 2	
Events	OPEN (n = 718)	ENDO (n = 716)	OPEN (n = 197)	ENDO (n = 199)
Above-ankle amputation				
N Event[a] (%)	74 (10)	106 (15)	30 (15)	28 (14)
4 years' KM event rate	0.14	0.19	0.25	0.19
5 years' KM event rate	0.15	0.19	–	0.25
Major reintervention				
N Event (%)	66 (9)	167 (23)	28 (14)	51 (26)
4 years' KM event rate	0.12	0.27	0.21	0.33
5 years' KM event rate	0.12	0.29	–	0.33
Death				
N Event (%)	237 (33)	269 (38)	51 (26)	48 (24)
4 years' KM event rate	0.38	0.43	0.44	0.45
5 years' KM event rate	0.45	0.51	0.44	0.56
MALE				
N Event (%)	305 (42)	409 (57)	84 (43)	95 (48)
4 years' KM event rate	0.51	0.66	0.61	0.68
5 years' KM event rate	0.58	0.71	–	0.72
Degradation of WIfI stage				
N Event (%)	124 (17)	180 (25)	35 (18)	42 (21)
4 years' KM event rate	0.23	0.31	0.27	0.33
5 years' KM event rate	0.23	0.32	–	0.38
Clinical failure				
N Event (%)	355 (49)	477 (67)	96 (49)	116 (58)
4 years' KM event rate	0.59	0.74	0.66	0.78
5 years' KM event rate	0.64	0.79	–	0.81

KM, Kaplan-Meier; *MALE*, major adverse limb events.
[a]Number of events while in study.

amputation, 313 (44%) had a major and/or minor reintervention to maintain index limb patency, 218 (30%) did not initially increase the ABI by 0.15 or TBI by 0.10, 226 (32%) had a subsequent decrease in the ABI of 0.15 or TBI of 0.10, and 113 (16%) had radiographic evidence of arterial stenosis or occlusion in the treated index limb (Table II).

In cohort 2, a total of 63% and 72% patients assigned to OPEN and ENDO, respectively experienced hemodynamic failure. Among those assigned to OPEN, 30 (15%) had an above-ankle amputation, 67 (34%) had a major and/or minor reintervention to maintain index limb patency, 54 (27%) did not initially increase the ABI by 0.15 or TBI by 0.10, 56 (28%) had a subsequent decrease in the ABI of 0.15 or TBI of 0.10, and 24 (12%) had radiographic evidence of arterial stenosis or occlusion in the treated index limb (Table II). Among those assigned to ENDO, 28 (14%) had an above-ankle amputation, 88 (44%) had a major and/or minor reintervention to maintain index limb patency, 64 (32%) did not initially increase the ABI by 0.15 or TBI by 0.10, 61 (31%) had a subsequent decrease in the ABI of 0.15 or TBI of 0.10, and 34 (17%) had radiographic evidence of arterial stenosis or occlusion in the treated index limb (Table II).

Resolution of presenting CLTI symptoms

In cohort 1, of the 718 and 716 patients assigned to OPEN and ENDO, respectively, 604 (84%) and 596 (83%) successfully resolved their initial CLTI symptoms. In cohort 2, of the 197 and 199 patients assigned to OPEN and ENDO, respectively, 159 (81%) and 159 (80%) successfully resolved their initial CLTI symptoms (Table III).

Incidence of new CLTI

In cohort 1, a total of 253 of the 603 (42%) patients assigned to OPEN, which resolved their presenting CLTI symptoms, developed new or recurrent CLTI, resulting in a person-years of follow-up incidence rate of 0.15. In comparison, 323 of the 617 (52%) patients assigned to ENDO, which resolved their presenting CLTI symptoms, developed new or recurrent CLTI, resulting in a person-years of follow-up incidence rate of 0.18 (Table IV).

In cohort 2, a total of 76 of the 171 (44%) patients assigned to OPEN, which resolved their presenting CLTI

Table II. Hemodynamic failure and component events for cohorts 1 and 2

	Cohort 1		Cohort 2	
Events	OPEN (n = 718)	ENDO (n = 716)	OPEN (n = 197)	ENDO (n = 199)
Above-ankle amputation				
N Event[a] (%)	74 (10)	106 (15)	30 (15)	28 (14)
4 years' KM event rate	0.14	0.19	0.25	0.19
5 years' KM event rate	0.15	0.19	–	0.25
Reintervention to maintain vascular patency in the index limb				
N Event (%)	223 (31)	313 (44)	67 (34)	88 (44)
4 years' KM event rate	0.39	0.53	0.57	0.54
5 years' KM event rate	0.41	0.54	–	–
Failure to increase ABI by at least 0.15 (for patients with noncompressible vessels, failure to increase the TBI by 0.10)				
N Event (%)	179 (25)	218 (30)	54 (27)	64 (32)
4 years' KM event rate	0.32	0.36	0.42	0.45
5 years' KM event rate	0.32	0.37	–	0.45
Decrease in ABI by 0.15 (or TBI drop of 0.10) or greater compared with postprocedure 30 days' value				
N Event (%)	165 (23)	226 (32)	56 (28)	61 (31)
4 years' KM event rate	0.31	0.39	0.41	0.45
5 years' KM event rate	0.31	0.4	–	–
Duplex ultrasound demonstrating occlusion of the graft or treated native vessel				
N Event (%)	63 (9)	–	17 (9)	–
4 years' KM event rate	0.13	–	0.14	–
5 years' KM event rate	0.13	–	–	–
Duplex ultrasound demonstrating critical graft stenosis				
N Event (%)	99 (14)	–	22 (11)	–
4 years' KM event rate	0.17	–	0.19	–
5 years' KM event rate	0.18	–	–	–
Angiogram demonstrating occlusion of any graft or treated vessel, or greater than 50% stenosis in the presence of recurrent clinical symptoms				
N Event (%)	76 (11)	113 (16)	24 (12)	34 (17)
4 years' KM event rate	0.15	0.2	0.27	0.23
5 years' KM event rate	0.17	0.21	–	0.34
Hemodynamic failure				
N Event (%)	438 (61)	523 (73)	124 (63)	144 (72)
4 years' KM event rate	0.75	0.82	0.82	0.83
5 years' KM event rate	0.75	0.83	–	–

ABI, Ankle brachial index; *KM*, Kaplan-Meier; *TBI*, toe brachial index.
[a]Number of events while on study.

symptoms, developed new or recurrent CLTI, resulting in a person-years of follow-up incidence rate of 0.23. In comparison, 83 of the 173 (48%) patients assigned to ENDO, which resolved their presenting CLTI symptoms, developed new or recurrent CLTI, resulting in a person-years of follow-up incidence rate of 0.23 (Table IV). Similar results were obtained after sensitivity analyses using imputed missing baseline characteristics.

Time to event analysis of clinical failure, hemodynamic failure, and resolution of presenting symptoms by ITT assignment. Time to event analyses for clinical failure, hemodynamic failure, and resolution of presenting symptoms within each cohort, by ITT assignment, are illustrated in the Fig. Multivariate analysis revealed that in cohort 1, there was a significant difference in time to clinical failure ($P < .001$), hemodynamic failure ($P <$

Table III. Resolution of initial critical limb-threatening ischemia (*CLTI*) symptoms in cohorts 1 and 2

Event	Cohort 1			Cohort 2		
	Overall (N = 1434)	OPEN (n = 718)	ENDO (n = 716)	Overall (N = 396)	OPEN (n = 197)	ENDO (n = 199)
CLTI resolution, No. (%)	1200 (84)	604 (84)	596 (83)	319 (81)	159 (81)	160 (80)

Table IV. Incidence of recurrent critical limb-threatening ischemia (*CLTI*) in cohorts 1 and 2

Event	Cohort 1				Cohort 2			
	OPEN (n = 603)	ENDO (n = 617)	IRR (95% CI)	*P* value	OPEN (n = 171)	ENDO (n = 173)	IRR (95% CI)	*P* value
Recurrent CLTI	253 (0.15)[a]	323 (0.18)[a]	0.81 (0.68, 0.95)	.01	76 (0.23)[a]	83 (0.23)[a]	0.91 (0.67, 1.25)	.57

CI, Confidence interval; *IRR*, incident rate ratio.
[a]Person-years of follow-up incidence rate.

.001), and resolution of presenting symptoms ($P = .009$) in favor of OPEN. This result was maintained even after adjustment for key baseline patient covariates (end-stage renal disease, prior revascularization, smoking, diabetes, age >80 years, WIfI stage, tissue loss, and IP disease). In cohort 2, there was a significantly lower rate of hemodynamic failure ($P = .006$) favoring OPEN, and no significant difference in time to clinical failure or resolution of presenting symptoms.

In both cohorts, assignment to OPEN was associated with a significantly reduced risk of clinical failure. In cohort 1, there was a 35% relative risk reduction (RRR) in clinical failure (hazard ratio [HR] [95% confidence interval (CI)]: 0.65 [0.56, 0.76], $P < .001$). In cohort 2, there was a 33% RRR in clinical failure (HR: 0.67 [0.50, 0.90], $P = .009$) (Table V). In both cohorts, assignment to OPEN was also associated with a significantly reduced risk of hemodynamic failure. In cohort 1, there was a 29% RRR in hemodynamic failure (HR: 0.71 [0.62, 0.82], $P < .001$), whereas in cohort 2, the risk reduction was 35% (HR: 0.65 [0.50, 0.85], $P < .001$) (Table V).

Randomization to OPEN further resulted in a 22% RRR in resolution of presenting CLTI symptoms in cohort 1 (HR: 1.22 [1.08, 1.38], $P = .002$). There was no significant difference in time to resolution of initial CLTI symptoms between the two arms in cohort 2. Comparable results were obtained when patients who required an above-ankle amputation before resolution of their initial symptoms were censored at the time of amputation, and when all-cause death and above-ankle amputation were treated as competing risks (Table V). The median time to initial symptom resolution in cohort 1 was 70.0 (95% CI: 58, 77) days for those assigned to OPEN and 91 (95% CI: 71, 99) days for those assigned to ENDO.

Incidence of new or recurrent CLTI

In cohort 1, randomization to OPEN resulted in a 19% reduction in risk of developing new or recurrent CLTI symptoms, reported as an incident rate ratio (HR: 0.81 [0.68, 0.95], $P < .01$). The incidence rates of new or recurrent CLTI symptoms in cohort 2 were not significantly different (Table II).

Multivariable analysis of clinical failure, hemodynamic failure, and development of new or recurrent CLTI symptom end points. We sought to identify factors associated with clinical failure, hemodynamic failure, and the development of new or recurrent CLTI symptoms by treatment assignment using Cox proportional hazards models for the full trial population (both arms and both cohorts combined [Table VI]; full models for the combined study population for these three end points are provided in Supplementary Table III, online only). For clinical failure, significant independent predictors included OPEN vs ENDO assignment (HR 0.75 [0.65, 0.87], $P < .001$), chronic kidney disease grade (stage 3 vs none: HR 1.40 [1.14, 1.71], $P = .001$; stage 4 vs none: HR 1.62 [1.09, 2.39], $P = .016$; dialysis-dependent vs none: HR 1.91 [1.53, 2.35], $P < .01$), rest pain vs tissue loss (HR 0.67 [0.53, 0.85], $P < .001$), and levels of disease treated (FP only vs FP+IP: HR 0.75 [0.65, 0.87], $P < .001$; IP vs FP+IP: HR 0.89 [0.67, 1.18], $P = .407$). Specific to the ENDO arm, patients who underwent isolated FP (HR 0.67 [0.49, 0.93], $P = .016$) or IP (HR 0.47 [0.29, 0.78], $P = .003$) intervention had a significantly reduced risk for clinical failure compared with those with combined FP+IP interventions. Those treated by plain balloon angioplasty were at a significantly higher risk of clinical failure (HR 1.72 [1.12, 2.65], $P = .013$) than those treated with a drug-eluting device (drug-coated balloon or drug-eluting stent) (Supplementary Table IV, online only). For OPEN surgery, chronic kidney disease grade (stage 4 vs none: HR 2.33 [1.00, 5.44], $P = .05$; dialysis-dependent vs none: HR 2.70 [1.70, 4.28], $P < .001$) and levels of disease treated (FP vs FP+IP: HR 0.63 [0.44, 0.90], $P = .01$; IP vs FP+IP: HR 0.57 [0.36, 0.92], $P = .02$) have a significant association with clinical failure.

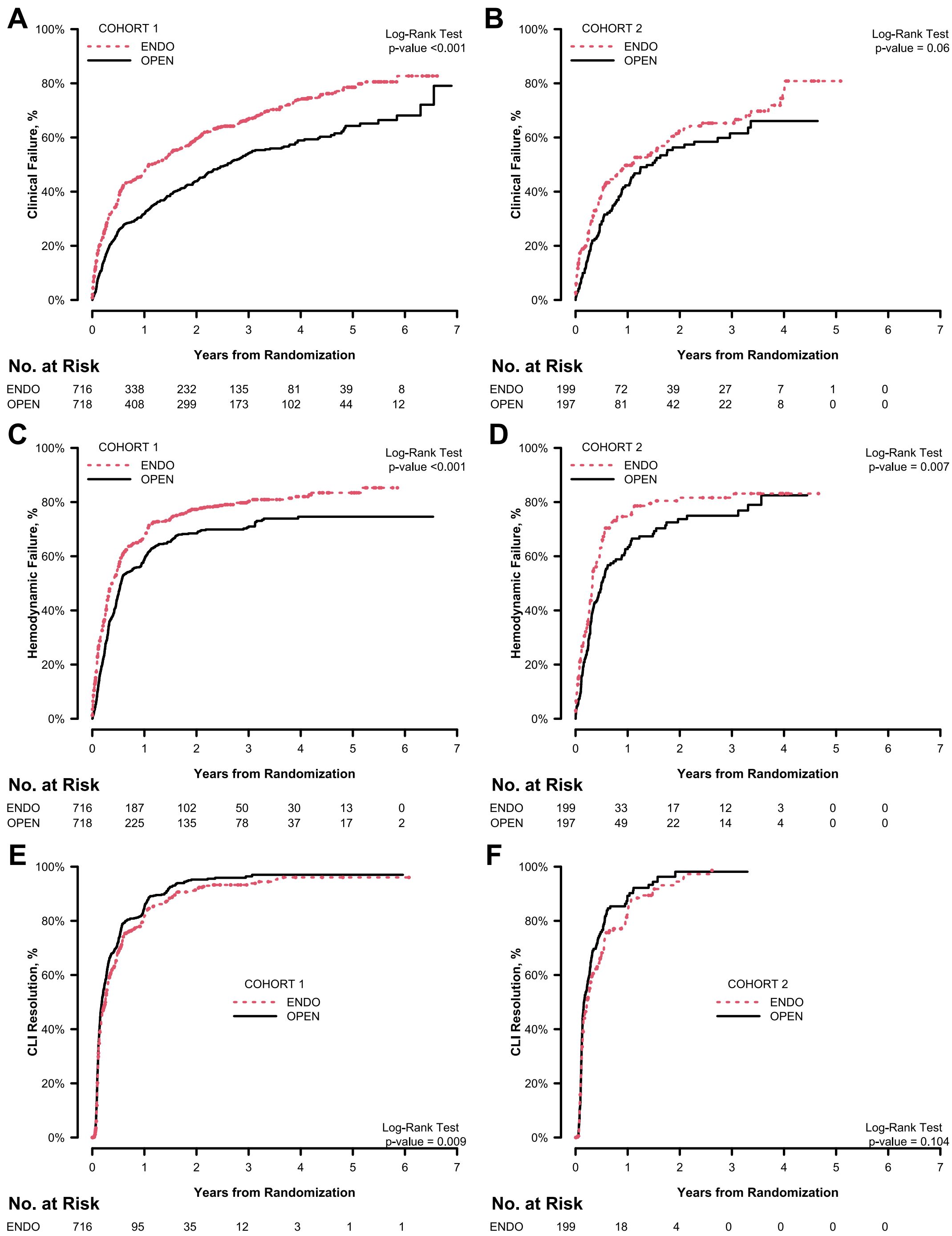

Fig. Time to clinical failure (**A** and **B**), hemodynamic failure **(C** and **D)**, and resolution of presenting CLTI symptoms **(E** and **F)** by ITT assignment in cohort 1 **(A**, **C**, **E)** and cohort 2 **(B**, **D**, **F)**. *CLTI*, Critical limb-threatening ischemia; *ITT*, intention-to-treat.

Factors associated with hemodynamic failure included OPEN vs ENDO assignment (HR 0.80 [0.65, 0.91], $P <$ .001), hyperlipidemia (HR 1.16 [1.00, 1.35], $P =$ .050), and isolated FP treatment vs FP+IP (HR 0.86 [0.75, 0.98], $P =$ .022, Table VI). There were no important predictors of hemodynamic failure within either the pooled ENDO or OPEN treatment arms.

Variables protective from the development of recurrent CLTI after initial resolution included OPEN vs ENDO assignment (HR 1.16 [1.03, 1.31], $P =$ .015), rest pain vs tissue loss (HR 1.94 [1.61, 2.33], $P <$ .001), and levels of disease treated (FP only vs FP+IP: HR 1.19 [1.05, 1.35], $P =$ .007), while advanced WIfI stage at presentation (stage 4 vs 1 or 2: HR 0.76 [0.64, 0.91], $P =$ .002)

Table V. Adjusted treatment effect estimates of clinical failure, hemodynamic failure, and resolution of initial critical limb-threatening ischemia (*CLTI*) by intention-to-treat (ITT) assignment

Parameter	Cohort 1				Cohort 2			
	OPEN (N = 603) N (%w event)	ENDO (N = 617) N (%w event)	HR[a] (95% CI)	*P* value	OPEN (N = 171) N (%w event)	ENDO (N = 173) N (%w event)	HR[a] (95% CI)	*P* value
Clinical failure	303 (50.25)	419 (67.91)	0.65 (0.56, 0.76)	<.001	81 (47.37)	104 (60.12)	0.67 (0.50, 0.90)	.009
Hemodynamic failure	377 (62.52)	464 (75.20)	0.71 (0.62, 0.82)	<.001	113 (65.50)	131 (75.14)	0.65 (0.50, 0.85)	.001
CLTI resolution								
From randomization to EOS	514 (85.24)	514 (83.31)	1.22 (1.08, 1.38)	.002	142 (83.04)	141 (81.50)	1.07 (0.84, 1.37)	.57
From randomization to EOS or AAAI[b]	490 (81.26)	472 (76.50)	1.23 (1.08, 1.40)	.002	134 (78.36)	133 (76.88)	1.08 (0.84, 1.39)	.55
From randomization to EOS with AAAI and death as competing risks	490 (81.26)	472 (76.50)	1.27 (1.12, 1.44)	<.001	134 (78.36)	133 (76.88)	1.13 (0.88, 1.43)	.34

AAAI, Above-ankle amputation of index leg; *CI*, confidence interval; *EOS*, end of study; *HR*, hazard ratio; *N (%w event)*, N (percentage with event).
[a]Adjusted for covariates prespecified in the statistical analysis plan: end-stage renal disease, prior infrainguinal revascularization procedure, smoking status, diabetes, age (>80 years), and baseline WIfI stage; excluding patients with missing baseline characteristics.
[b]Patients having AAAI without resolving presenting CLTI symptoms, censored at the time of amputation.

independently predicted recurrent disease (Table VI). In the ENDO arm, only rest pain vs tissue loss reduced the risk of recurrent symptoms (HR 2.64 [1.68, 4.15, $P < .001$), whereas in the OPEN arm, both higher WIfI stage (3 vs 1 or 2: HR 0.63 [0.46, 0.86], $P = .003$); 4 vs 1 or 2: HR 0.49 [0.36, 0.66], $P < .001$) and multilevel disease (FP vs FP+IP: HR 1.45 [1.09, 1.92], $P = .011$) increased the risk of recurrent symptoms (Supplementary Table IV, online only).

DISCUSSION

Durable clinical and hemodynamic benefit after revascularization for CLTI is imperative to both resolve symptoms and prevent recurrence, reinterventions, and limb loss. To date, well-designed end points that reliably capture important elements of clinical or hemodynamic success in CLTI are lacking. To address this unmet need, we sought to examine several novel end points that have as their central focus the deterioration over time of the clinical and hemodynamic benefit conferred by revascularization. These end points are concordant with suggestions on trial design made in the SVS Objective Performance Goals for CLTI.[1] There was notable consistency in the findings of these planned secondary analyses, where initial treatment with open surgical bypass, particularly among patients with adequate SSGSV (cohort 1), was associated with significantly improved clinical and hemodynamic durability, faster resolution of initial CLTI symptoms, and greater protection against recurrent CLTI symptoms.

WIfI incorporates the three major factors that directly impact amputation risk, the status of the wound, the level of ischemia, and the degree of infection.[5] Clinical failure as defined here shares similarity to the primary end point of MALE-free survival, but additionally captures deterioration in limb status as represented by WIfI stage. This confers the ability to assess the impact of the initial treatment strategy longitudinally at the level of the foot, and specifically to influence the key drivers of wound healing and recovery from CLTI. This represents a significant advance as it moves beyond the blunt metrics of amputation and all-inclusive death and better reflects the durable impact of revascularization on the limb and the patient.

Our findings of reduced clinical failure with open surgical bypass are driven in large part by lower rates of reintervention[6] and major amputations. The advantages of surgical revascularization in patients with more advanced disease/WIfI status have been well documented.[7,8] Factors such as conduit availability for bypass, advanced ischemia, and renal failure are key considerations in planning revascularization procedures. Our findings that renal failure, tissue loss compared with rest pain, and multisegment disease were associated with earlier clinical failure in both arms combined are also in line with prior reports[9,10] of interest; renal failure conferred particular risk to patients undergoing open surgery. Although one may have predicted that bypass in the setting of multisegment disease would better protect against clinical failure than endovascular treatment, a similar effect was seen with both. Our results also demonstrated a reduced risk of clinical failure with the use of drug-eluting balloons over untreated balloons within the endovascular arm, consistent with a number of published randomized controlled trials on drug-coated balloons.[11]

A primary goal of limb revascularization in CLTI is improved perfusion to resolve ischemic symptoms.

Table VI. Multivariate Cox regression analysis of factors independently associated with clinical failure, hemodynamic failure, and recurrent critical limb-threatening ischemia (*CLTI*) for the full trial population (both arms and both cohorts combined)

End point	Comparison	HR (95% CI)	*P* value
Clinical failure	OPEN vs ENDO	0.75 (0.65, 0.87)	<.001
	CKD grade		
	Stage 3 vs no CKD	1.40 (1.14, 1.71)	.001
	Stage 4 vs no CKD	1.62 (1.09, 2.39)	.016
	Dialysis vs no CKD	1.91 (1.55, 2.35)	<.001
	Rest pain vs tissue loss	0.67 (0.53, 0.85)	<.001
	Level treated[a]		
	FP vs FP+IP	0.75 (0.65, 0.87)	<.001
	IP vs FP+IP	0.89 (0.67, 1.18)	.047
Hemodynamic failure	OPEN vs ENDO	0.80 (0.70, 0.91)	<.001
	Hyperlipidemia	1.16 (1.00, 1.35)	.050
	Level treated[a]		
	FP vs FP+IP	0.86 (0.75, 0.98)	.022
	IP vs FP+IP	1.20 (0.92, 1.56)	.177
Recurrent CLTI	OPEN vs ENDO	1.16 (1.03, 1.31)	.015
	Rest pain vs tissue loss	1.94 (1.61, 2.33)	<.001
	Level treated[a]		
	FP vs FP+IP	1.19 (1.05, 1.35)	.007
	IP vs FP+IP	0.96 (0.72, 1.26)	.747
	WIfI stage		
	Stage 3 vs stage 1 or 2	0.89 (0.75, 1.06)	.209
	Stage 4 vs stage 1 or 2	0.76 (0.64, 0.91)	.002

CI, Confidence interval; *CKD*, chronic kidney disease; *HR*, hazard ratio.
[a]Levels treated: FP = femoropopliteal only; IP = infrapopliteal only; FP+IP = both.

Similar to clinical failure, the use of hemodynamic failure as a BEST-CLI trial end point is a novel effort to more precisely ascertain the comparative hemodynamic impact of the treatment received and measure its durability. Inclusive of the macro elements of amputation and interventions (both major and minor) undertaken to maintain patency, this end point also incorporates changes in ankle and toe brachial indices, as well as imaging evidence of stenosis or occlusion in the index graft or treated arterial segment. This composite end point definition sets a threshold for "failure" based on a combination of clinical, anatomic, and noninvasive measures that have been used in defining technical success and restenosis. As such, the results reported here lack historical context. Given the inherent hazards of recoil, intimal hyperplastic restenosis, and progressive atherosclerosis in this high-risk CLTI population,[12] treatment-specific hemodynamic longevity is of paramount importance as a determinant of the quality of that intervention. Because clinical failure and hemodynamic failure may reflect both patient-specific and procedure-specific variables, we analyzed both within the overall trial population and then separately for the ENDO and OPEN arms. Apart from randomization to ENDO, predictors of hemodynamic failure were multisegment disease and hyperlipidemia within the combined cohort. There were no associations seen in either of the individual treatment arms.

Marked by impaired healing, a substantial reintervention burden, a high risk of recurrence, and the need for diligent long-term surveillance, CLTI is often compared with cancer. One could argue that the most valuable measure of revascularization success is the ability to resolve presenting CLTI symptoms and prevent the development of new or recurrent symptoms, in essence maintaining a state of "remission." As such, our third novel end point tracks resolution of initial symptomatology and the time it takes to do so. Finally, CLTI recurrence captures incidence rates of recrudescent rest pain and/or tissue loss as well as new ischemic lesions. We believe that this report represents the first such documentation of the burden of recurrent disease in a large CLTI cohort. Over the entire course of follow-up, more than 40% of patients who initially resolved their presenting symptoms developed recurrent CLTI. This striking finding highlights both the need for better secondary preventive care and the need for greater durability in limb revascularization in this population. On multivariate analysis, randomization to OPEN, rest pain vs tissue loss, WIfI stage 1 or 2 vs stage 4, and treatment of single segment disease all conferred protection from recurrent disease. Within the OPEN arm, higher WIfI stage negatively impacted recurrent disease, while tissue loss vs rest pain on presentation was a very strong predictor of recurrent CLTI symptoms in those randomized to endovascular treatment.

The limitations inherent to the pragmatic randomized controlled study design of BEST-CLI and the unknown impact of adherence to recommended medical therapy have been discussed elsewhere.[2,4] Of note, the trial was not powered for the planned analyses of the novel end points described herein, conferring a risk of type II errors. Standardized protocols for either reinterventions or follow-up surveillance were not mandated, introducing the risk of ascertainment bias. Discordant rates of missingness between the two study arms, particularly with regard to postprocedure imaging and hemodynamic metrics, may have further influenced the results seen. Wound status was investigator reported, as available resources did not allow for camera confirmation. We also do not yet know the degree to which specific surgical or endovascular techniques affected these secondary end points or their subcomponents. Finally, when designing BEST-CLI, we believed the lack of an agreed-upon standard for defining and tracking patency following endovascular treatment precluded inclusion of this as an end point.

Given the lack of historical precedent for these novel measures, they remain to be validated. Notwithstanding these limitations, the innovative end points reported herein capture a measure of granularity not represented by amputation rates alone and not present in the main end point of the BEST-CLI, MALE plus all-cause death, or in amputation-free survival, the main end point of BASIL-2.[13] As such, they aspire to provide additional highly relevant context with which to better understand the overall trial results and the specific downstream effects of our treatment efforts, and in so doing further guide informed decision-making for both patients and providers. The soberingly high rate of recurrent symptoms seen and the fact that two-thirds of patients experienced some form of treatment failure in the manner defined highlight the challenges inherent to CLTI care and the opportunities for improved effectiveness and durability of revascularization in this population. Analysis of the presenting anatomic patterns of the randomized patients, currently underway, should provide additional help toward understanding the degree of disease complexity and determining the generalizability of the trial findings.

CONCLUSIONS

Patients randomized to open surgical bypass in the BEST-CLI trial had more sustained benefit from their index revascularization as reflected in lower rates of both clinical failure and hemodynamic failure. This finding was seen regardless of the type of conduit used. In the setting of available saphenous vein, initial treatment with open surgery also resulted in more effective and quicker resolution of the presenting symptoms of CLTI and, equally as important, in preventing the burden of recurrent symptoms. These robust findings, derived from a set of novel end points that should be further developed and used, provide valuable context for the primary results of BEST-CLI and further underscore the impact (and limitations) of the initial revascularization strategy on long-term outcomes for patients with CLTI.

AUTHOR CONTRIBUTIONS

Conception and design: MM, AF, MD, TH, KR, MC
Analysis and interpretation: MM, AF, GD, KM, EC, LC, AK, PS, BH, MD, TH, MS, KR, MC
Data collection: MM, AF, KM, EC, LC, AK, PS, BH, MD, TH, MS, KR, MC
Writing the article: MM
Critical revision of the article: MM, AF, GD, KM, EC, LC, AK, PS, BH, MD, TH, MS, KR, MC
Final approval of the article: MM, AF, GD, KM, EC, LC, AK, PS, BH, MD, TH, MS, KR, MC
Statistical analysis: GD, TH
Obtained funding: MM, AF, MD, MS, KR
Overall responsibility: MM, MC

FUNDING

BEST-CLI was supported by the National Heart, Lung, and Blood Institute of the National Institutes of Health under Award Numbers U01HL107407, U01HL107352, and U01HL115662. The following entities also provided funding to the BEST-CLI trial during the follow-up period (2019-2021). These include Physician Societies: Vascular InterVentional Advances (VIVA), Society for Vascular Surgery, New England Society for Vascular Surgery, Western Vascular Society, Eastern Vascular Society, Midwest Vascular Surgery Society, Southern Association of Vascular Surgeons, Canadian Society for Vascular Surgery, Society for Clinical Vascular Surgery, Society of Interventional Radiology, Vascular and Endovascular Surgery Society, Society for Vascular Medicine; and Industry sources: Janssen, Gore, Becton Dickinson and Company, Medtronic, Cook, Boston Scientific, Abbott, Cordis, Cardiovascular Systems Inc. As of September 1, 2022, ongoing BEST-CLI research is funded primarily by the Novo Nordisk Foundation.

DISCLOSURES

M.T.M. is the advisor for Janssen. A.F. grants from Novo Nordisk Foundation; is a consultant for Sanifit, LeMaitre, and BioGenCell; and is on the advisor board of Dialysis-X and iThera Medical. L.C.C. is a consulting speaker for Cook Medical. P.A.S. is a consultant for Medtronic, Boston

Scientific, Philips, Silk Road, Surmodics, Cagent, and Limflow. B.M.H. reports research grants (site PI for clinical trials, no compensation) from Boston Scientific and NIH/NHLBI. M.D.D. is a consultant for Cook Medical, W.L. Gore, and Boston Scientific. K.R. receives income as a consultant or member of a scientific advisory board for the following entities: Abbott Vascular, Althea Medical, Angiodynamics, Auxetics, Becton-Dickinson, Boston Scientific, Contego, Crossliner, Innova Vascular, Inspire MD, Janssen/Johnson and Johnson, Magneto, Mayo Clinic, MedAlliance, Medtronic, Neptune Medical, Penumbra, Philips, Surmodics, Terumo, Thrombolex, Truvic, Vasorum, and Vumedi; and owns equity or stock options in the following entities: Access Vascular, Aerami, Althea Medical, Auxetics, Contego, Crossliner, Cruzar Systems, Endospan, Imperative Care/Truvic, Innova Vascular, InspireMD, JanaCare, Magneto, MedAlliance, Neptune Medical, Orchestra, Prosomnus, Shockwave, Skydance, Summa Therapeutics, Thrombolex, Vasorum, and Vumedi. K.R. or his institution (on his behalf) receives research grants from the following entities: NIH, Abiomed, Boston Scientific, Novo Nordisk Foundation, Penumbra, and Gettinge-Atrium. He serves as a member of the Board of Directors of the following organization: The National PERT Consortium. M.S.C. has a relationship with Abbott Vascular DSMB. The remaining authors report no conflicts.

REFERENCES

1. Conte MS, Geraghty PJ, Bradbury AW, et al. Suggested objective performance goals and clinical trial design for evaluating catheter-based treatment of critical limb ischemia. *J Vasc Surg*. 2009;50:1462–1473.
2. Menard MT, Farber A, Assmann SF, et al. Design and rationale of the best endovascular versus best surgical therapy for patients with critical limb ischemia (BEST-CLI) trial. *J Am Heart Assoc*. 2016;5:e003219.
3. Bonaca MP, Bauersachs RM, Anand SS, et al. Rivaroxaban in peripheral artery disease after revascularization. *N Engl J Med*. 2020;382:1994–2004.
4. Farber A, Menard MT, Conte MS, et al. Surgery or endovascular therapy for chronic limb-threatening ischemia. *N Engl J Med*. 2022;387:2305–2316.
5. Mills JL, Conte MS, Armstrong DG, et al. The society of vascular surgery lower extremity threatened limb classification system: risk stratification based on wound, ischemia, and foot infection (WIfI). *J Vasc Surg*. 2014;59:200–234.
6. Conte MS, Azene A, Doros G, et al. Secondary interventions following open versus endovascular revascularization for chronic limb threatening ischemia in the BEST-CLI trial. *J Vasc Surg*. 2024;79:1428–1437.e4.
7. Shirasu T, Takagi H, Gregg A, et al. Predictability of the global limb anatomic staging system (glass) for technical and limb related outcomes: a systematic review and meta-analysis. *Eur J Vasc Endovasc Surg*. 2022;64:32–40.
8. Meecham L, Popplewell M, Bate G, et al. Evaluation of the Global Limb Anatomic Staging System in patients with chronic limb-threatening ischemia undergoing endovascular intervention for femoropopliteal disease. *J Vasc Surg*. 2023;77:474–479.e473.
9. Meltzer AJ, Evangelisti G, Graham AR, et al. Determinants of outcome after endovascular therapy for critical limb ischemia with tissue loss. *Ann Vasc Surg*. 2014;28:144–151.
10. Vierthaler L, Callas PW, Goodney PP, et al. Determinants of survival and major amputation after peripheral endovascular intervention for critical limb ischemia. *J Vasc Surg*. 2015;62:655–664.
11. Aboyans V, Ricco J-B, Bartelink M-LEL, et al. 2017 ESC guidelines on the diagnosis and treatment of peripheral arterial diseases, in collaboration with the European Society for Vascular Surgery (ESVS): document covering atherosclerotic disease of extracranial carotid and vertebral, mesenteric, renal, upper and lower extremity arteries. *Eur Heart J*. 2018;39:763–816.
12. Sorber R, Dun C, Kawai Q, Black JH, Makary MA, Hicks CW. Early peripheral vascular interventions for claudication are associated with higher rates of late interventions and progression to chronic limb threatening ischemia. *J Vasc Surg*. 2023;77:1720–1731.
13. Bradbury AW, Moakes CA, Popplewell M, et al. A vein bypass first versus best endovascular treatment first revascularization strategy for patients with chronic limb threatening ischaemia who required an infra-popliteal, with or without an additional more proximal infra-inguinal revascularization procedure to restore limb perfusion (BASIL-2): an open label, randomised, multicentre, phase 3 trial. *Lancet*. 2023;401:1798–1809.

Submitted Apr 13, 2024; accepted Jul 21, 2024.

Additional material for this article may be found online at www.jvascsurg.org.

Supplementary Table I (online only). Baseline patient characteristics

Characteristics	Cohort 1			Cohort 2		
	Overall (N = 1434)	Bypass (n = 718)	ENDO (n = 716)	Overall (N = 396)	Bypass (n = 197)	ENDO (n = 199)
Mean age ± SD, years	66.9 ± 9.9	66.9 ± 9.8	67.0 ± 10.0	68.6 ± 9.2	68.4 ± 8.8	68.8 ± 9.6
Female sex, No. (%)	408 (28.5)	201 (28.0)	207 (28.9)	111 (28.0)	56 (28.4)	55 (27.6)
Hispanic, No./total No. (%)	187/1433 (13.0)	82/717 (11.4)	105/716 (14.7)	53 (13.4)	28 (14.2)	25 (12.6)
Race, No./total No. (%)[a]						
White	1028/1423 (72.2)	500/711 (70.3)	528/712 (74.2)	275/390 (70.5)	143/194 (73.7)	132/196 (67.3)
Black	275/1423 (19.3)	156/711 (21.9)	119/712 (16.7)	96/390 (24.6)	40/194 (20.6)	56/196 (28.6)
Other	120/1423 (8.4)	55/711 (7.7)	65/712 (9.1)	19/390 (4.9)	11/194 (5.7)	8/196 (4.1)
Hypertension, No./total No. (%)	1238/1424 (86.9)	620/712 (87.1)	618/712 (86.8)	350/395 (88.6)	171/196 (87.2)	179/199 (89.9)
Hyperlipidemia, No./total No. (%)	1041/1423 (73.2)	521/712 (73.2)	520/711 (73.1)	299/395 (75.7)	147/196 (75.0)	152/199 (76.4)
Coronary artery disease, No./total No. (%)	617/1424 (43.3)	301/712 (42.3)	316/712 (44.4)	204/395 (51.6)	97/196 (49.5)	107/199 (53.8)
Congestive heart failure, No./total No. (%)	79/1422 (5.6)	38/711 (5.3)	41/711 (5.8)	27/395 (6.8)	12/196 (6.1)	15/199 (7.5)
Chronic obstructive pulmonary disease, No./total No. (%)	208/1424 (14.6)	100/712 (14.0)	108/712 (15.2)	69/395 (17.5)	34/196 (17.3)	35/199 (17.6)
History of stroke, No./total No. (%)	190/1424 (13.3)	91/712 (12.8)	99/712 (13.9)	62/395 (15.7)	38/196 (19.4)	24/199 (12.1)
End-stage kidney disease, No./total No. (%)	151/1423 (10.6)	67/712 (9.4)	84/711 (11.8)	45/395 (11.4)	25/196 (12.8)	20/199 (10.1)
Current smoking, No./total No. (%)	509/1424 (35.7)	264/712 (37.1)	245/712 (34.4)	140/395 (35.4)	69/196 (35.2)	71/199 (35.7)
Diabetes, No./total No. (%)	1023/1424 (71.8)	513/712 (72.1)	510/712 (71.6)	238/395 (60.3)	122/196 (62.2)	116/199 (58.3)
Medications, No./total No. (%)						
Treated pharmacologically for smoking	97/1424 (6.8)	49/712 (6.9)	48/712 (6.7)	26/395 (6.6)	11/196 (5.6)	15/199 (7.5)
At least one statin	1001/1424 (70.3)	503/713 (70.5)	498/711 (70.0)	307/394 (77.9)	153/195 (78.5)	154/199 (77.4)
At least one antiplatelet drug	1025/1424 (72.0)	508/713 (71.2)	517/711 (72.7)	303/394 (76.9)	153/195 (78.5)	150/199 (75.4)
Previous infrainguinal revascularization on index limb, No./total No. (%)	77/1423 (5.4)	40/711 (5.6)	37/712 (5.2)	40/393 (10.2)	20/194 (10.3)	20/199 (10.1)
Randomization stratum, No./total No. (%)						
Ischemic rest pain without significant infrapopliteal disease	127 (8.9)	64 (8.9)	63 (8.8)	49 (12.4)	24 (12.2)	25 (12.6)
Tissue loss without significant infrapopliteal disease	348 (24.3)	175 (24.4)	173 (24.2)	95 (24.0)	48 (24.4)	47 (23.6)
Ischemic rest pain with significant infrapopliteal disease	164 (11.4)	82 (11.4)	82 (11.5)	70 (17.7)	35 (17.8)	35 (17.6)
Tissue loss with significant infrapopliteal disease	795 (55.4)	397 (55.3)	398 (55.6)	182 (46.0)	90 (45.7)	92 (46.2)
SVS WIfI stage, No./total No. (%)						
Stage 1	84/1223 (6.9)	46/606 (7.6)	38/617 (6.2)	15/345 (4.3)	9/172 (5.2)	6/173 (3.5)
Stage 2	321/1223 (26.2)	162/606 (26.7)	159/617 (25.8)	127/345 (36.8)	62/172 (36.0)	65/173 (37.6)
Stage 3	370/1223 (30.3)	167/606 (27.6)	203/617 (32.9)	94/345 (27.2)	48/172 (27.9)	46/173 (26.6)
Stage 4	448/1223 (36.6)	231/606 (38.1)	217/617 (35.2)	109/345 (31.6)	53/172 (30.8)	56/173 (32.4)
Mean index leg ABI ± SD[a]	0.58 ± 0.32	0.58 ± 0.31	0.59 ± 0.34	0.54 ± 0.30	0.53 ± 0.27	0.54 ± 0.32
Mean ankle pressure ± SD, mm Hg[a]	84.9 ± 47.7	85.2 ± 46.2	84.5 ± 49.2	81.3 ± 49.6	80.4 ± 47.3	82.2 ± 51.8
Mean toe pressure ± SD, mm Hg[a,b]	36.3 ± 25.7	36.5 ± 27.7	36.1 ± 23.5	31.0 ± 21.7	37.0 ± 23.5	25.5 ± 18.4

ABI, Ankle brachial index; *SD*, standard deviation; *SVS*, Society for Vascular Surgery.
[a]ABI was missing on 463 subjects in cohort 1 (224 bypass and 239 ENDO) and 116 in cohort 2 (62 bypass and 54 ENDO), body mass index was missing on 67 subjects in cohort 1 (36 bypass and 31 ENDO) and 19 in cohort 2 (9 bypass and 10 ENDO), Ankle pressure was missing on 426 subjects in cohort 1 (205 bypass and 221 ENDO) and 108 in cohort 2 (57 bypass and 51 ENDO), and Toe pressure was missing on 826 subjects in cohort 1 (418 bypass and 408 ENDO) and 227 in cohort 2 (117 bypass and 110 ENDO).
[b]In cohort 1, race was significantly associated with treatment group (*P* = .04), and in cohort 2, toe pressure was significantly different between treatment groups (*P* = .002).

Supplementary Table II (online only). Subtypes of major and minor reinterventions performed

	Cohort 1			Cohort 2		
Reintervention type	**Overall**	**OPEN**	**ENDO**	**Overall**	**OPEN**	**ENDO**
Major reinterventions						
All major	365	113	252	135	56	79
New bypass graft placement	219	33	186	68	14	54
Open surgical thrombectomy	35	17	18	24	12	12
Mechanical thrombectomy	42	19	23	17	13	4
Jump/interposition graft revision	28	21	7	8	5	3
Pharmacologic thrombolysis	54	30	24	25	13	12
Minor reinterventions						
All minor	563	261	302	148	70	78
Surgical patch angioplasty	40	19	21	15	4	11
Balloon angioplasty	504	247	257	132	64	68
Atherectomy	46	13	33	10	4	6
Laser treatment	8	0	8	4	0	4
Stent placement	146	48	98	53	21	32
Stent-graft placement	24	6	18	16	5	11

Supplementary Table III (online only). Multivariable Cox regression analysis of factors associated with clinical failure, hemodynamic failure, and recurrent critical limb-threatening ischemia (*CLTI*) for the full study population (full model, all covariates examined)

Covariate (effect)	HR (95% CI), *P* value
Multivariable Cox regression model for clinical failure	
Randomized surgery	$P < .001$
OPEN vs ENDO	0.75 (0.65, 0.87), $P < .001$
CKD grade	$P < .001$
Stage 3 vs no CKD	1.40 (1.14, 1.71), $P = .001$
Stage 4 vs no CKD	1.62 (1.09, 2.39), $P = .016$
Dialysis dependent vs no CKD	1.91 (1.55, 2.35), $P < .001$
Diabetes	$P = .30$
Yes vs no	1.10 (0.92, 1.30), $P = .297$
Tissue loss vs rest pain	$P < .001$
Rest pain vs tissue loss	0.67 (0.53, 0.85), $P = .001$
Hyperlipidemia	$P = .16$
Yes vs no	1.13 (0.95, 1.33), $P = .158$
Level treated (combination of OPEN and ENDO)	$P < .001$
FP vs FP+IP	0.75 (0.65, 0.87), $P < .001$
IP vs FP+IP	0.89 (0.67, 1.18), $P = .407$
Smoking history	$P = .36$
Prior (>1 year) vs never	1.12 (0.92, 1.36), $P = .251$
Prior (2 weeks to 1 year) vs never	1.24 (0.93, 1.64), $P = .143$
Current vs never	1.03 (0.84, 1.26), $P = .766$
WIfI stage	$P = .64$
3 vs 1 or 2	0.94 (0.76, 1.16), $P = .557$
4 vs 1 or 2	0.90 (0.73, 1.12), $P = .349$
Multivariable Cox regression model for hemodynamic failure	
Randomized surgery	$P = .001$
OPEN vs ENDO	0.80 (0.70, 0.91), $P = .001$
CKD grade	$P = .07$
Stage 3 vs no CKD	1.03 (0.84, 1.25), $P = .781$
Stage 4 vs no CKD	0.91 (0.58, 1.42), $P = .675$
Dialysis dependent vs no CKD	1.33 (1.07, 1.64), $P = .009$
Diabetes	$P = .89$
Yes vs no	0.99 (0.85, 1.15), $P = .888$
Tissue loss vs rest pain	$P = .16$
Rest pain vs tissue loss	0.86 (0.70, 1.06), $P = .164$
Hyperlipidemia	$P = .05$
Yes vs no	1.16 (1.00, 1.35), $P = .054$
Level treated (combination of OPEN and ENDO)	$P = .01$
FP vs FP+IP	0.86 (0.75, 0.98), $P = .022$
IP vs FP+IP	1.20 (0.92, 1.56), $P = .177$
Smoking history	$P = .41$
Prior (>1 year) vs never	1.12 (0.93, 1.33), $P = .229$
Prior (2 weeks to 1 year) vs never	1.11 (0.85, 1.45), $P = .429$
Current vs never	1.00 (0.83, 1.20), $P = .985$
WIfI stage	$P = .16$
3 vs 1 or 2	1.00 (0.82, 1.22), $P = .974$
4 vs 1 or 2	1.15 (0.95, 1.40), $P = .152$
Multivariable Cox regression model for recurrent CLTI	
Randomized surgery	$P = .02$
OPEN vs ENDO	1.16 (1.03, 1.31), $P = .015$
CKD grade	$P = .26$
Stage 3 vs no CKD	0.90 (0.75, 1.08), $P = .259$
Stage 4 vs no CKD	1.22 (0.83, 1.80), $P = .310$
Dialysis dependent vs no CKD	0.86 (0.69, 1.08), $P = .198$
Diabetes	$P = .09$
Yes vs no	0.89 (0.77, 1.02), $P = .094$
Tissue loss vs rest pain	$P < .001$
Rest pain vs tissue loss	1.94 (1.61, 2.33), $P = .000$
Hyperlipidemia	$P = .19$
Yes vs no	0.91 (0.79, 1.05), $P = .188$
Level treated (combination of OPEN and ENDO)	$P = .02$
FP vs FP+IP	1.19 (1.05, 1.35), $P = .007$
IP vs FP+IP	0.96 (0.72, 1.26), $P = .747$
Smoking history	$P = .40$
Prior (>1 year) vs never	1.14 (0.95, 1.35), $P = .153$
Prior (2 weeks to 1 year) vs never	1.18 (0.93, 1.52), $P = .179$
Current vs never	1.06 (0.89, 1.27), $P = .489$
WIfI stage	$P = .01$
3 vs 1 or 2	0.89 (0.75, 1.06), $P = .209$
4 vs 1 or 2	0.76 (0.64, 0.91), $P = .002$

CI, Confidence interval; *CKD*, chronic kidney disease; *FP*, femoropopliteal; *HR*, hazard ratio; *IP*, infrapopliteal.

Supplementary Table IV (online only). Multivariable Cox regression analysis of factors associated with secondary end points within each of the intention-to-treat (*ITT*) assigned treatment arms

End point	Covariate	HR (95% CI)	*P* value
OPEN			
Clinical failure	CKD grade		
	Stage 3 vs no CKD	1.51 (0.96, 2.35)	.072
	Stage 4 vs no CKD	2.33 (1.00, 5.44)	.050
	Dialysis vs no CKD	2.69 (1.69, 4.28)	<.001
	Level OPEN treated		
	FP vs FP+IP	0.63 (0.44, 0.90)	.012
	IP vs FP+IP	0.57 (0.36, 0.92)	.022
Recurrent CLTI	Level OPEN treated		
	FP vs FP+IP	1.45 (1.09, 1.92)	.011
	IP vs FP+IP	1.16 (0.80, 1.69)	.422
	WIfI stage		
	Stage 3 vs stage 1 or 2	0.63 (0.46, 0.86)	.003
	Stage 4 vs stage 1 or 2	0.49 (0.36, 0.66)	<.001
ENDO			
Clinical failure	Level treated		
	FP vs FP+IP	0.67 (0.49, 0.93)	.016
	IP vs FP+IP	0.47 (0.29, 0.78)	.003
	Type of intervention		
	Bare metal vs drug bal	1.03 (0.73, 1.45)	.884
	Angio only vs drug bal	1.72 (1.12, 2.65)	.013
Recurrent CLTI	Rest pain vs tissue loss	2.64 (1.68, 4.15)	<.001

CI, Confidence interval; *CKD*, chronic kidney disease; *CLTI*, critical limb-threatening ischemia; *FP*, femoropopliteal; *HR*, hazard ratio; *IP*, infrapopliteal.